

From HM-Side Adaptive K to Variable- K RTL: A CHIA Loop for HEVC RDO Co-Design

Nguyen Hieu Duong and Duy Hieu Bui
Vietnam National University

Corresponding author: 24020497@vnu.edu.vn

Artifact: <https://github.com/duongnguyenhieu/CHIA-RDO>, release chia-rdo-v1.0.0

Abstract—HEVC intra coding exposes 35 luma prediction modes, making full rate-distortion optimization (RDO) expensive. CHIA-RDO separates the resulting co-design problem into two explicit implementations. Patched HM-16.20 software ranks all 35 modes and selects $K \in \{4, 16, 35\}$ from normalized rough cost; the RTL receives that already-ranked prefix and does not implement pruning, rough mode decision, or CABAC bit estimation. Its hardware contribution is a variable- K batch scheduler, 16 serial 4×4 candidate-evaluation lanes, and stable balanced winner reduction. The frozen HM-side policy reduces held-out candidate RDO evaluations by 38.21% while retaining 97.91% of Full-RDO winners; aggregate Y-PSNR BD-rate is -0.026% on the limited corpus. Physical feedback replaces monolithic candidate logic and a linear winner scan, yielding a routed kernel at a derived 68.120 MHz and 8.134 M candidates/s, $2.128\times$ the P1 baseline proxy. A CHIA loop preserves the software contract while independently searching and verifying this RTL microarchitecture.

Index Terms—Agentic co-design, adaptive candidate selection, FPGA, HEVC, rate-distortion optimization, RTL.

I. INTRODUCTION AND BACKGROUND

HIGH Efficiency Video Coding (HEVC) improves compression by evaluating a large set of spatial prediction choices, but this flexibility makes intra-mode search computationally expensive. For each prediction unit, the HM reference encoder first applies rough mode decision (RMD), using SATD and an estimated signaling cost to rank 35 luma modes. It then performs the more expensive rate-distortion optimization (RDO), which reconstructs and compares shortlisted candidates using distortion and coded-rate estimates [3].

Fast intra-decision methods reduce this workload by pruning candidates before full RDO. Fixed-size shortlists are simple but cannot adapt to content: smooth or strongly directional blocks may need few candidates, whereas textured or ambiguous blocks benefit from broader search. Prior adaptive methods use features such as relative SATD and statistical risk to trade encoding effort for coding efficiency [2]. These methods are usually evaluated as software algorithms, without considering how each candidate budget maps to a parallel hardware scheduler.

That mapping creates a cross-layer problem. An FPGA with P lanes executes $\lceil K/P \rceil$ batches, so two policies with equal mean K can incur different latency and lane utilization. Conversely, increasing P can reduce batches while increasing area, routing pressure, and critical-path delay. Algorithm selection and RTL architecture therefore need a shared evidence loop rather than independent optimization.

CHIA-RDO addresses this gap without claiming that the pruning policy itself is hardware. It connects software coding quality, cycle-accurate RTL correctness, and routed physical feedback while preserving a precise boundary between them. Its principal contributions are:

- 1) a frozen *HM-side* Adaptive- K policy and matched coding curves;
- 2) an explicit event contract carrying the selected K , stable rank, HM bit counts, QP, lambda, references, and original samples into RTL;
- 3) a variable- K RTL architecture whose scheduler, serial candidate lanes, and balanced stable-argmin differ from fixed-size or cost-only candidate arrays; and
- 4) a CHIA graph preserving RTL tests, routed feedback, cache behavior, and Pareto history for replay and claim auditing.

II. HM-SIDE ADAPTIVE- K ALGORITHM

Software responsibility. Every operation in this section executes in patched HM-16.20, not in SystemVerilog. The policy treats a low normalized best rough cost as evidence that one mode already predicts the block well, so a short expensive-RDO prefix is sufficient; ambiguous blocks retain more candidates. For each luma PU, HM:

- 1) computes $C[m] = \text{SATD}[m] + \text{bits}[m]\sqrt{\lambda}$ for every mode $m = 0, \dots, 34$;
- 2) stable-sorts modes by increasing C , retaining lower mode ID on an exact tie;
- 3) computes

$$\hat{C}_{\min} = \frac{\min_m C[m]}{WH(2^b - 1)}, \quad (1)$$

for PU size $W \times H$ and luma bit depth b ;

- 4) selects

$$K = \begin{cases} 4, & \hat{C}_{\min} \leq 0.04, \\ 16, & 0.04 < \hat{C}_{\min} \leq 0.20, \\ 35, & \hat{C}_{\min} > 0.20; \end{cases} \quad (2)$$

- 5) runs candidate RDO on exactly the first K ranked modes, retaining the earlier stable rank on an exact RD-cost tie.

Difference from other selection rules. Fixed- K always evaluates the same prefix. Our relative-SATD baseline instead counts modes within a relative SATD threshold and rounds that count to an allowed K . The selected rule uses only the absolute

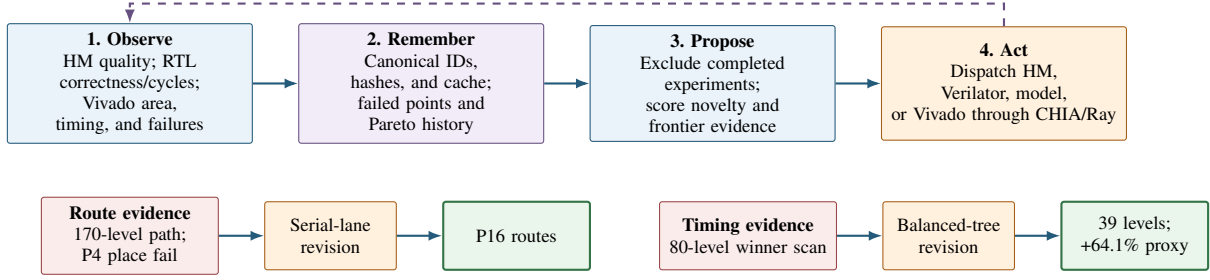


Fig. 1. The history-aware agentic loop and two concrete evidence-to-action traces. The controller proposes experiments; each versioned RTL revision then re-enters the same verification and backend loop.

best signaling-aware rough cost, normalized by PU area and sample range; it needs neither a trained Bayesian classifier nor subset statistics. It is not a reproduction of Gwon–Choi [2]. A bounded discovery comparison selected $(0.04, 0.20; 4, 16, 35)$ below a predeclared 1% BD-rate gate; its identity was then frozen before held-out evaluation.

For coding experiments, HM executes candidate RDO on the first K ranks. For RTL regression, HM additionally exports each candidate’s integer bit count and floating-point lambda; the vector bridge rounds lambda to Q16.16 and packages an event. The policy deliberately excludes lane count P , scheduler state, QP, and activity. Hardware latency follows only afterward as $\lceil K/P \rceil$ batches, so RTL exploration cannot change the selected prefix. This artifact is a software-to-vector/test flow, not a live HM–FPGA integration. Matched curves use pinned HM-16.20, Main 8-bit 4:2:0 All-Intra, four frames, and QP 22/27/32/37.

III. END-TO-END CO-DESIGN ARCHITECTURE

Here *agentic* denotes a closed observe–remember–propose–act loop, not an LLM that emits unchecked RTL. Native ChiaFunction nodes dispatch tools asynchronously. Typed results expose quality, correctness, cycle, resource, timing, and failure signals; atomic records preserve canonical experiment IDs, source hashes, successful and failed points, and Pareto history. The deterministic proposal policy removes completed IDs, rewards underexplored configurations and frontier boundaries, and gives promoted RTL measurements priority. Track A rejects any mutation of the frozen Adaptive- K identity.

Actions remain auditable: the controller selects and dispatches experiments, while each source-level microarchitectural revision is versioned before it re-enters regression and routing. Thus Fig. 1 does not claim autonomous RTL generation; it shows how remembered backend evidence changes the next design hypothesis. The campaign accumulated 114 history records, 107 successful logical experiments, 11 matched HM curves, and a 10-point Pareto set. A calibrated graph then screened 832 P /depth/width combinations and produced 16 Pareto points; replay returned 832/832 cache hits without rerunning tools. The HM branch measures policy quality, whereas Verilator/Vivado evaluate only the contracted kernel, so agentic exploration cannot move Adaptive- K across the boundary in Fig. 2.

IV. HARDWARE ALGORITHM AND RTL ARCHITECTURE

Not implemented in RTL: 35-mode RMD, stable sorting, the threshold test that chooses K , lambda derivation, and CABAC-context bit estimation. Implemented in RTL: variable- K dispatch, complete 4×4 luma candidate arithmetic for each supplied rank, fixed-point RD cost, and stable winner selection. This distinction is the central hardware/software contract.

A start transaction supplies K ; up to 35 ranked 6-bit modes and aligned 24-bit rate counts; 6-bit QP; 32-bit λ ; seventeen 8-bit top/left references; and the row-major 4×4 8-bit original luma block. The final `full_rdo_pway_optimized` elaboration uses $P = 16$, depth 3, and a 31-bit cost. Its ready/valid scheduler dispatches candidate $bP + l$ to lane l and masks the partial final batch.

The scheduler differs from fixed- K hardware by accepting any event count up to 35, preserving input rank, and masking the last partial batch. It differs from a cost-only array because each lane computes prediction through SSE rather than receiving distortion. Within a lane, one sample or coefficient

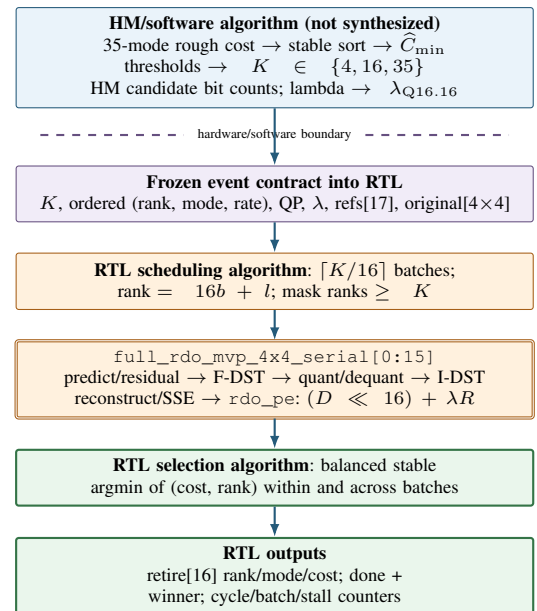


Fig. 2. Algorithm partition and selected RTL. Adaptive- K ends above the boundary; hardware preserves its ranked prefix while evaluating candidates and selecting the winner. Each serial lane takes 132 cycles per candidate.

TABLE I
CODING RESULTS RELATIVE TO THE FULL-RDO ANCHOR

Method / Corpus	Candidate Rule	Avg. K	RDO Red.	Retention	BD-Rate
Full-RDO anchor	All 35 rough-ranked modes	35.000	0.00%	100.00%	0.000%
Fixed- K 8 / discovery	First 8 ranked modes	8.000	77.14%	68.05%	+3.574%
Relative-SATD / discovery	Threshold count, quantized K	25.624	26.79%	96.07%	+0.528%
Frozen threshold / discovery	Normalized rough cost; $K = 4/16/35$	18.651	46.71%	90.25%	+0.755%
Fixed- K 16 / held-out	First 16 ranked modes	16.000	54.29%	85.70%	+0.016%
Frozen threshold / held-out	Same policy, no retuning	21.628	38.21%	97.91%	-0.026%
HW-aware Track-B / P8	Joint quality/cycle objective	16.456	52.98%	88.02%	-0.054%

RDO reduction is $1 - \text{Avg. } K/35$; retention is the fraction of blocks whose Full-RDO winner remains in the prefix. Negative aggregate BD-rate on three synthetic held-out families is not a general quality claim.

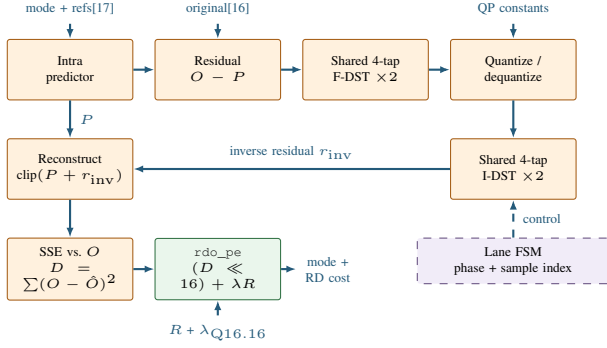


Fig. 3. Candidate datapath inside one serial RTL lane. This is bounded HEVC 4×4 arithmetic, not the Adaptive- K policy. Registers between phases store 16 intermediate samples while arithmetic is reused across positions; 16 identical lanes feed the stable reduction tree in Fig. 2.

is advanced per FSM cycle, replacing the earlier monolithic combinational candidate path. Across lanes, a balanced tree replaces the earlier 16-way linear winner scan. The comparison is lexicographic: lower cost wins, then lower supplied rank. Per-lane retirement exposes rank/mode/cost; event completion also returns the winner and cycle, batch, stall, and lane-activity counters. A no-backpressure batch retires in 134 cycles.

A. Verification Evidence

Final campaigns executed 121,600 vector/candidate checks. The depth-3 serial corpus alone contains 8,960 vectors covering all 35 modes and four QPs; the tree campaign adds 10,560 P1/P8/P16 evaluations including backpressure. Stage values, order, winners, cycle counts, dropped/duplicate checks, and interfaces all passed with zero failures.

V. SOFTWARE-ALGORITHM AND RTL EVALUATION

The direct held-out comparison exposes the software-algorithm tradeoff: Fixed- K 16 prunes more work, but loses the Full-RDO winner on 14.30% of blocks; Adaptive- K spends extra candidates on ambiguous content and lowers that loss to 2.09%. It removes only 13.95% and 13.30% of evaluations on mixed-texture and repetitive families, but 87.36% on smooth/edge content. Track-B is separately versioned and does not alter the frozen policy.

The Gwon-Choi relative-SATD Bayesian method reports 31.54% average encoding time reduction and +0.93% BD-rate

against HM-16.6 under its own Intra Main condition [2]. It starts from stock-HM RMD modes and includes a minimum-risk Bayesian classifier. Our relative-SATD baseline instead starts from all 35 modes and uses threshold counting only, so the published result is context rather than a direct comparison.

VI. RTL EVOLUTION FROM PHYSICAL FEEDBACK

The first P-way baseline replicated a monolithic candidate datapath. At P1, the critical path traversed prediction, residual, both forward-transform passes, quant/dequant, both inverse passes, reconstruction, and distortion: 170 logic levels, including 69 CARRY8 levels. P2 reached 175 levels, and P4 exceeded LUT/CARRY capacity before placement.

The next proposal serialized work within each lane and inserted state boundaries between major arithmetic phases. This raised Fmax and reduced per-lane area, allowing P16 to route, but a serial 16-way winner scan became an 80-level path. The final balanced tree reduced that path; the routed critical path is now 39 levels inside one lane's prediction/residual datapath and ends at a residual-sample RAM input.

A. Measured Interpretation

To compare routed variants under a common 5 ns constraint, we report

$$f_{\text{derived}} = \frac{1}{5 \text{ ns} - \text{WNS}}, \quad T_{\text{kernel}} = \frac{P f_{\text{derived}}}{C_{\text{batch}}}. \quad (3)$$

This post-route comparison proxy is distinct from encoder frames/s. The selected P16 design routes with $\text{WNS} = -9.680$ ns, hence $f_{\text{derived}} = 68.120$ MHz and $T_{\text{kernel}} = 8.134$ M candidates/s for 134-cycle batches.

B. Resource Budget

On the KV260 device (117,120 LUTs, 1,248 DSPs), the final design uses 52.26% of LUTs and 65.38% of DSPs. Against the project's 70%-of-device integration budget, this is 74.66% of the LUT allowance and 93.47% of the DSP allowance, leaving 57 DSPs. P16 is the throughput endpoint of the Pareto frontier, not the minimum-area or minimum-power point.

TABLE II
ROUTED IMPLEMENTATIONS

Architecture	<i>P</i>	LUT/DSP	WNS	MHz	M/s
Mono. combinational	1	36,454/178	−60.395	15.292	3.823
Mono. combinational	2	70,817/356	−64.653	14.357	7.178
Serial d3, w31	8	30,813/408	−9.620	68.399	4.084
Serial P16 scan	16	60,232/816	−19.092	41.508	4.956
Serial P16 tree	16	61,207/816	−9.680	68.120	8.134

All rows routed. WNS is in ns; MHz is derived from the routed critical path; M/s is the candidate-kernel rate.

VII. ROUTED MICROARCHITECTURE COMPARISON

The balanced tree costs 975 LUTs and 45 FFs over the P16 serial scan, leaves DSP count unchanged, and reduces estimated power from 2.338 W to 2.288 W. It improves derived Fmax by 64.1% and kernel rate by 64.1%. Relative to the monolithic P1 proxy, its candidate rate is 2.128 $\times$; relative to P2 it is 1.133 $\times$. The calibrated CHIA model predicts P16 within 0.14% for LUT, 0.45% for Fmax/throughput, and 0.43% for power.

VIII. REPRODUCIBILITY AND DELIVERABLES

Local/GCP parity produced identical bitstream, reconstruction, and trace hashes for a reference encode. GCP workers later replayed RTL campaigns with zero failures. Post-run Compute Engine queries found no remaining instances, disks, or reserved addresses. Actual billed spend and promotional-credit balance remain unavailable and are not replaced by estimates.

The artifact preserves policy versions, manifests, source and corpus hashes, raw Vivado reports, machine-readable result schemas, and Pareto history. Principal deliverables are CHIA proposal/evaluation graphs and the deterministic policy controller; pinned HM patches, matched quality curves, and synthetic fixtures; parameterized SystemVerilog, golden vectors, and Verilator campaigns; and analytical replay, model calibration, routed reports, and parsers.

IX. LIMITATIONS AND NEXT STEP

The coding study uses small deterministic synthetic All-Intra workloads, not standard natural-video classes. RTL is

limited to 4 $\times$ 4 8-bit luma and excludes chroma, inter prediction, larger transforms, RDOQ, syntax coding, and hardware CABAC/context evolution. Exact HM candidate bit counts are supplied externally. The 31-bit cost width is corpus-qualified, not formally bounded for arbitrary input. Most importantly, this artifact does not connect a running HM encoder directly to the FPGA: software selects and packages each event, and RTL begins only at the interface in Fig. 2.

Resource scale is also a limitation. The selected design uses 816 DSP48E2 primitives, exactly 51 per lane times 16 lanes. “Serial” describes candidate execution within a lane; it does not imply one shared multiplier. Runtime products in prediction, quant/dequant, squared error, and the 24 $\times$ 32-bit λR term infer multiple DSP slices, and the RTL does not explicitly share these multipliers across lanes. The P16 scan already uses 60,232 LUTs; the balanced tree adds 975 LUTs and 45 FFs but no DSPs, producing the 61,207-LUT total. A final hierarchical utilization split by operation was not preserved, so finer DSP attribution would require a new synthesis report.

All physical runs are out of context and omit board-level clocks, pins, AXI, and encoder integration. The routed result is therefore reported using its derived critical-path frequency. Future work will pipeline forward-transform accumulation and repeat regression, routing, and board-level integration.

Within these limits, CHIA-RDO demonstrates the intended agentic co-design behavior: it preserves a software-quality contract, searches reproducibly, checks a bit-exact kernel, and uses failed backend evidence to produce a specific microarchitectural revision rather than merely requesting more parallelism.

REFERENCES

- [1] CHIA, “An open framework for agile and principled hardware/software co-design research,” arXiv:2606.27350, 2026.
- [2] D. Gwon and H. Choi, “Relative SATD-based minimum risk Bayesian framework for fast intra decision of HEVC,” *KSII Trans. Internet Inf. Syst.*, vol. 13, no. 1, pp. 385–405, Jan. 2019, doi: 10.3837/tiis.2019.01.022.
- [3] ITU-T/ISO/IEC, “High Efficiency Video Coding Test Model HM-16.20,”
- [4] CHIA-RDO artifact. [Online]. Available: <https://github.com/duongnguyenhieu/CHIA-RDO>, release chia-rdo-v1.0.0.

AI assistance was used in preparing this artifact.